

Bioassay Suite — GUI Refresh

Implementation guide for Claude Code. Presentation-layer + small UI-restructure pass.

No business logic, server reactives, analysis pipeline, or report-token CSS in Rmd templates change.

MOCKUP

Bioassay Suite — GUI refresh.html

TOKENS

tokens.css

SCOPE

Style guide · 5 wizard screens · Report output

MODE

Visual + minor UI restructure

§ 1 Design system

1.1 Type

Add Google Fonts to `app.R` UI head, or self-host into `www/fonts/`:

```
tags$head(
  tags$link(rel = "preconnect", href = "https://fonts.googleapis.com"),
  tags$link(rel = "preconnect", href = "https://fonts.gstatic.com", crossorigin = NA),
  tags$link(rel = "stylesheet",
    href = "https://fonts.googleapis.com/css2?
family=IBM+Plex+Mono:wght@400;500;600&family=IBM+Plex+Sans:wght@400;500;600&family=IBM+Plex+Seri
f:wght@400;500&display=swap"),
  tags$link(rel = "stylesheet", type = "text/css", href = "style.css")
)
```

Stack:

- **IBM Plex Sans** — UI body, labels, buttons, headings
- **IBM Plex Mono** — all numbers, scientific values, hex codes, file names, kbd-like tokens
- **IBM Plex Serif** — only the report's display title (*Bioassay Analysis Report*)

Type scale (matches `tokens.css`):

ROLE	SIZE	WEIGHT
Display (report title)	28–36 px	500 (serif)
H1 / page title	22 px	500
H2 / section	17 px	600
H3	14 px	600
Body	13 px	400
Small / help	12 px	400
Micro / mono labels	11 px	500

1.2 Color (Okabe-Ito, colorblind-safe)

All semantic colors map to Okabe-Ito so reds/greens never collide for CVD users.

ROLE	HEX	SOFT BG	NOTES
 PASS	#009e73	#e6f5ee	bluish-green
 WARN	#e69f00	#fbf0d9	orange
 FAIL	#d55e00	#fbe6d9	vermillion
 INFO	#0072b2	#dfecf5	blue
 ACCENT	#1f3a5f	#e8edf4	deep navy — active tab, primary buttons, links

Neutrals: `--c-bg #f7f8f9`, `--c-surface #fff`, `--c-surface-2 #f1f3f5`, `--c-line #e3e6ea`, ink `#0f1722`, ink-3 `#5b6573`.

Replace the current `style.css` color usage:

- `#1976D2` → `var(--c-accent) (#1f3a5f)`
- `#388E3C` → `var(--c-pass) (#009e73)`
- `#FFC107` / `#FFA000` warn boxes → `var(--c-warn)` + `--c-warn-soft`
- Plotly heatmap palette → use Okabe-Ito stops `#1f3a5f` → `#0072b2` → `#009e73` → `#f0e442` (perceptually uniform-ish, CB-safe). Dose-response point colors use Okabe-Ito blue / orange / vermillion.

1.3 Geometry & shadows

- Radii: `3 / 6 / 10 px` (`--r-1/2/3`); pills `999 px`.
- Shadows: hairline only (`var(--sh-1)`). Drop the heavy `box-shadow: 0 4px 12px rgba(0,0,0,0.15)` hover translate on quick-start cards.
- Spacing scale: `4 / 8 / 12 / 16 / 24 / 32 / 48 px`.

1.4 Drop-in CSS

Replace the entirety of `www/style.css` with the contents of `tokens.css` from this project, plus the small mapping rules for Shiny / Bootstrap selectors (see § 7).

§ 2 App shell — top bar & stepper

2.1 Top bar

Replace the current `uiOutput("app_title_ui") + br() + actionButton("start_tour", ...)` block with a single horizontal top bar:

- **Left:** monogram tile (26×26, `--c-accent` background, white "B") + wordmark *Bioassay Suite* + version pill `v0.9.0` (mono, ink-4).
- **Right:** quiet *Guided tour*, *Docs*, divider, then the language `<select>` styled as a small button.

2.2 Numbered stepper (replaces `nav-pills`)

Keep `tabsetPanel(id = "wizard_tabs", type = "pills", ...)` server-side, but in CSS restyle the pills into a **numbered stepper**:

- Each pill becomes a row of: 22 px circle (number, mono) + label (12 px).
- **Done:** filled navy circle with check icon.
- **Current:** filled navy circle + soft halo (`box-shadow: 0 0 0 3px var(--c-accent-soft)`) + 2 px navy underline.
- **Upcoming:** outlined circle, ink-3 label.
- **Connectors:** 12 px hairline between steps.

Purely CSS — no server changes. Selectors:

```
.nav-pills > li > a { /* stepper row styling */ }
.nav-pills > li.active > a { /* current */ }
```

If Bootstrap's `<a>` markup makes the circle hard, wrap the tab labels server-side: `tabPanel("1. Configuration", ...) → tabPanel(HTML('<span class="num">01</span><span class="label">Configuration</span>'), ...)`.

§ 3 Tab 1 — Configuration

3.1 Quick Start — collapse to 3 tiles

Current: gradient pink/blue panel, 2 rows × 3 columns of buttons (Instant demo / Configure manually).

Replace with a **single card** containing **three preset tiles**. The instant-demo / manual split is removed from the surface — clicking a tile loads the preset; demo data ships with *RBA Saxitoxin* and *ELISA Cortisol* automatically (apps already have it in `examples/`); *ELISA Custom* is the blank template.

```
# Server: the 6 buttons collapse to 3.
# Map each tile click to the existing handler:
qs_rba_stx      <- observeEvent(input$qs_rba_stx,      { ... existing demo path ... })
qs_elisa_cortisol <- observeEvent(input$qs_elisa_cortisol, { ... existing demo path ... })
qs_elisa_custom  <- observeEvent(input$qs_elisa_custom, { ... existing manual path ... })
```

If you want to keep the "configure manually without demo data" option for STX/Cortisol, expose it via a small `<details>` "More options" disclosure under the tiles, not as a duplicated row.

Tile markup pattern (each tile = button card with a 4 px left rail in Okabe-Ito blue / bluish-green / purple):

```
<button class="bs-card preset-tile">
  <div class="rail rail-blue"></div>
  <div class="body">
    <div class="title">RBA · Saxitoxin</div>
    <div class="sub">8 standards, triplicate · demo data included</div>
  </div>
  <svg class="ico-arrow">...</svg>
</button>
```

Delete the gradient background `linear-gradient(135deg, #E3F2FD 0%, #F3E5F5 100%)`.

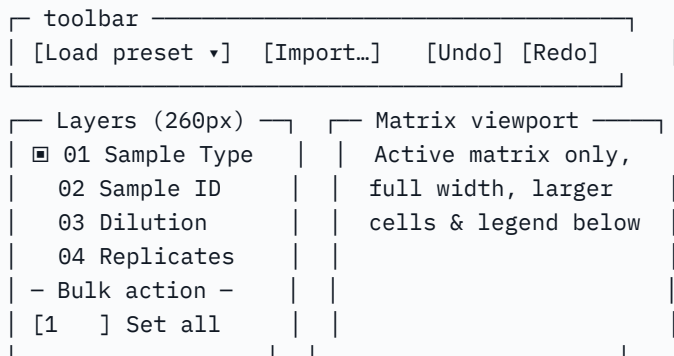
3.2 Assay & Standards

- Reorganize as two two-column rows: left = section heading + 1-line description; right = card with the inputs. Drop the wellPanel blue background.
- Standard inputs: 4-column grid, each input mono font, with unit suffix in the label `S1 (pg/mL)`.
- Replace the conditionalPanel "Toxin standard used:" UI with the same 2-col pattern.

§ 4 Tab 2 — Plate Layout (2×2 → layer switcher)

Biggest functional change. All four matrices stay in the data model and reactives; only one renders at a time.

4.1 New layout



4.2 R / Shiny implementation

The four `rHandsontableOutput`s already exist: `matrix_type`, `matrix_id`, `matrix_dilution`, `matrix_replicate`.

Do not delete them. Wrap them in conditional panels driven by a new `radioButtons("active_layer", ...)` (rendered as the styled tablist):

```
radioButtons("active_layer", NULL,
             choices = c("Sample Type" = "type",
                         "Sample ID"   = "id",
                         "Dilution"   = "dilution",
                         "Replicates"  = "rep"),
             selected = "type")

conditionalPanel("input.active_layer == 'type'",   div(id = "matrix_type_section", ...)),
conditionalPanel("input.active_layer == 'id'",     div(id = "matrix_id_section", ...)),
conditionalPanel("input.active_layer == 'dilution'", div(id = "matrix_dilution_section", ...)),
conditionalPanel("input.active_layer == 'rep'",    div(id = "matrix_replicate_section", ...))
```

The `radioButtons` markup is then CSS-targeted to render as the vertical tablist:

```
#active_layer .radio { /* row */ }
#active_layer input[type="radio"]:checked + span { /* active state */ }
```

Or replace with `actionButton`s + a `reactiveVal("active_layer")` if radio styling is awkward. The handsontable widgets keep all their existing observers, change-handlers, and `shared$matrix_*` reactive state. **Switching layers is just toggling visibility – no rebuild.**

4.3 Bulk action panel

Move the existing "Set all wells to fraction:" + Apply controls from the dilution-matrix toolbar into the **bottom of the sidebar**. It should adapt to the active layer:

- **Type:** dropdown of types (Standard / Sample / Blank / NSB / B0)

- **ID:** text input (prefix)
- **Dilution:** numeric (current behavior)
- **Rep:** text input (group code)

Existing `apply_uniform_dilution` and `apply_uniform_extraction` observers stay; add equivalents for the other layers if useful, or keep this Phase 2.

4.4 Legend & helper text

Below the active matrix, render a 5-chip color legend (Standard / Sample / Blank / NSB / B0). The Dilution layer keeps the existing "Enter the fraction of original sample strength remaining..." help block — restyle as an info card (`var(--c-info-soft)` + `var(--c-info)` border, no yellow).

4.5 Tissue-weight section (ELISA)

Stays. Restyle the section header and bulk-action row to match the same toolbar pattern. *No structural change.*

4.6 QC fields (RBA only)

Stay. Move below the matrix viewport, in a 520 px max-width card with two side-by-side inputs (QC concentration · Expected Hill slope). No yellow.

§ 5 Tabs 3 – 5

5.1 Tab 3 — Upload & Preview

- Two-column layout: left = file picker + heatmap, right = import summary card + raw-data sample card.
- File row: 32×32 file-icon tile (accent-soft bg) + name + size + *Replace* + green PASS pill.
- **Heatmap:** swap the current Plotly `RdBu` / `Spectral` palette for an Okabe-Ito-derived ramp (navy → blue → bluish-green → yellow). In `plotly` / `scales`, set:

```
scale_fill_gradientn(colors = c("#1f3a5f", "#0072b2", "#009e73", "#f0e442"))
```

- Import summary becomes a key/value table (label · mono value), not a green wellPanel.

5.2 Tab 4 — Analysis Settings

- Single max-width 880 px card.
- Each setting is a row: 260 px label + sub-description on the left, controls on the right, hairline divider between rows.
- Replace the yellow "Advanced Options" wellPanel with the same row pattern — the visual hierarchy is enough; no need for a special background.

- All `checkboxGroupInput` / `radioButtons` keep their IDs and choices.

5.3 Tab 5 — Generate Report

- Two columns: left = Output card + Pre-flight card stacked; right = Run notes card + Last run summary.
- Format checkboxes become **format tiles**: the active tile gets navy border + accent-soft background; an `<input type="checkbox">` is still inside (so server input bindings remain). Same for language pills.
- Pre-flight items: keep the existing list and the `aria-live` polite region. Each item = colored chip-glyph + description + small mono meta. Use `is-pass` (green check), `is-warn` (orange `!`), `is-fail` (vermillion `x`).
- Big "Generate Report" button: `bs-btn is-primary is-block`, navy, full width inside the card foot.

§ 6 Report output (`reports/unified_analysis_template.Rmd`)

6.1 Title block

Use IBM Plex Serif for the H1 only; everything else stays sans. Remove the duplicate "ELISA Analysis Report" subheading — the title + run metadata line replaces it.

```
<div class="report-title">
  <div class="eyebrow">Analysis report · run NN of YYYY-MM-DD</div>
  <h1 class="serif">Bioassay Analysis Report</h1>
  <div class="meta">Assay · Date · Analyst · Software</div>
</div>
```

6.2 KPI strip (new)

Insert a 5-tile KPI strip immediately under the title, **before** the existing Executive Summary box:

```
[ R² · curve fit ] [ RMSE ] [ LLOQ ] [ ULOQ ] [ • Overall ]
```

Each tile: small mono uppercase label, large mono value (22 px, weight 500), small mono unit. The status tile uses a colored dot (`var(--c-pass/warn/fail)`) and the word.

Pull values from the existing `model_stats` list / Rmd chunk that already produces R², RMSE, LLOQ, ULOQ, and overall pass/fail. Just render an extra HTML block above the existing summary.

6.3 Executive summary

Keep the textual block. Restyle as a quiet card (1 px line, white bg) — drop the blue tinted background and rounded corner border. Two-column key/value layout with ink-3 muted labels.

6.4 Section numbering

Section headings adopt a `§ N` prefix in mono ink-4 above the H2. Optional but matches the IDE feel.

6.5 Dose-response curve

- Switch ggplot theme to `theme_minimal()` with `panel.grid.minor = element_blank()` and gridline color `#e3e6ea`.
- Standards: filled black circles (Okabe-Ito).
- Samples in range: `#0072b2` (blue) circles.
- `>ULOQ` : `#d55e00` (vermillion) up-triangles.
- `<LLOQ` : `#e69f00` (orange) down-triangles.
- Shade the EC20–EC80 quantification band with `#0072b2` at alpha 0.05 + dashed bounds at alpha 0.5.

6.6 Results table

Replace the `kableExtra` default with these rules:

- mono numerics, right-aligned, tabular numerals (`font-variant-numeric: tabular-nums`)
- header row: small caps mono, `--c-surface-2` background, ink-3 color
- alternating row hover (`--c-surface-2`)
- Status column uses the same pill component (`is-pass` interpolated · `is-warn` extrapolated)

6.7 ELISA collapsible sections

Style the `<details><summary>` to match: chevron + sentence-case heading, no aggressive blue.

§ 7 File-by-file checklist

FILE	CHANGE
<code>www/style.css</code>	Replace contents with <code>tokens.css</code> from the mockup project + Shiny mapping rules below.
<code>app.R</code>	Add Plex font <code><link></code> to <code>tags\$head</code> . Restructure top bar (<code>#language_toggle_section</code> + <code>start_tour</code> → unified top bar div). Replace the <code>quickstart_section</code> 6-button block with 3 tiles. In Tab 2, wrap each <code>matrix_*_section</code> in a <code>conditionalPanel</code> driven by new <code>radioButtons("active_layer", ...)</code> .
<code>server_layout.R</code>	Add observer for <code>input\$active_layer</code> (no-op on data; just controls visibility). Optionally extend <code>apply_uniform_dilution</code> pattern to <code>apply_uniform_type / _id / _rep</code> .
<code>server_config.R</code>	Collapse <code>qs_rba_stx_demo</code> + <code>qs_rba_stx_manual</code> handlers into a single <code>qs_rba_stx</code> (and same for cortisol / custom).
<code>i18n.R</code>	Translation keys for the 3 collapsed Quick Start tiles + <i>Layers / Set all</i> / KPI labels (R^2 , RMSE, LLOQ, ULOQ, Overall).
<code>reports/unified_analysis_template.Rmd</code>	Add KPI strip block before the existing summary. Update <code><style></code> block / external CSS link. Switch ggplot palette + theme. Restyle kable tables with the rules in § 6.6.
<code>reports/plot_functions.R</code>	Replace any hard-coded <code>RdBu</code> / <code>Spectral</code> / <code>viridis</code> palette with the Okabe-Ito stops in § 1.2. Keep <code>viridis</code> as fallback if a continuous palette is needed.
<code>reports/report_functions.R</code>	No change.

Shiny / Bootstrap CSS overrides to append to `tokens.css`

```
/* Map Bootstrap 3 (Shiny default) selectors to design tokens */
.nav-pills > li > a      { /* stepper row — see §2.2 */ }
.btn-primary            { /* alias var(--c-accent) */ }
.btn-success            { background: var(--c-pass); border-color: var(--c-pass); }
.well                  { background: var(--c-surface); border: 1px solid var(--c-line); }
box-shadow: var(--sh-1); }
.form-control           { /* alias .bs-input */ }
.handsontable th        { background: var(--c-surface-2) !important; color: var(--c-ink-3) !important; }
.handsontable td         { font-family: var(--ff-mono); font-size: 11px !important; }
.shiny-input-radiogroup .radio { /* if used for layer switcher */ }
```

§ 8 Out of scope (do not change)

- `drc` curve fitting, model fallback chain (LL.4 → LL.3 → log-linear)
 - Auto-save behavior (60 s)
 - Outlier-detection logic
 - Tissue-normalization math
 - Multi-wavelength concordance analysis
 - Bilingual translation engine — only **add** keys, never rewire
 - Output folder naming convention (`YYYY-MM-DD_NN`)
 - `runApp()` / `runGitHub()` entry points
 - File import logic (`utils_import_v3.R` , `utils_import_multiwavelength.R`)
-

§ 9 Acceptance checklist

- No console errors when running `shiny::runApp("app.R")`
 - All 5 tabs accessible; pre-flight panel still validates
 - Quick Start: 3 tiles, all 3 paths reach the right step (RBA Sax → demo data loaded; ELISA Cort → demo loaded; ELISA Custom → blank template)
 - Plate Layout: switching the layer toggle does not lose edits in inactive matrices (verify by editing each, switching away and back)
 - Heatmap renders with Okabe-Ito-derived gradient
 - Report KPI strip shows numbers from `model_stats`
 - Run a CB-safety check: simulate deuteranopia / protanopia (e.g. Chrome DevTools rendering > emulate vision deficiency); pass/warn/fail pills + curve markers remain distinguishable
 - All translation keys present in EN and ES
 - `.napkin` / RStudio / browser print of report still produces clean PDF
-

§ 10 Reference

Visual mockup file: `Bioassay Suite – GUI refresh.html` (open in a browser; pan/zoom canvas; click any artboard label to focus fullscreen).

Token CSS source: `tokens.css`.